

Lecture 11 Exponential family of distributions, Completeness.

Defn A random variable with probability distr (pmf or pdf) f_θ has a distr belonging to the 2-parameter exponential family distr, if f_θ can be expressed in the following manner:

$$f_\theta(x) = h(x) \exp\left(\sum_{i=1}^s \eta_i(\theta) T_i(x) - B(\theta)\right), \theta \in \Theta. \longrightarrow (*)$$

Example Practically all common type of distr, discrete or continuous, belong to the exp. family of distr, like, Binomial, Poisson, Geometric, Normal, Gamma, Beta, chi-square, log-Normal, Exponential (scale, only), etc.

Note Distr for which the support depends on any of the parameters do not belong to the exponential family, e.g., $U(0, \theta)$, $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, $U(\theta_1, \theta_2)$, Exponential with location, Shifted Gamma, etc.
In fact, any generalisation of inclusion of a location parameter would violate the defn.

Ex 1 $X \sim \text{Gamma}(\alpha, \beta)$, $f_\theta(x) = \frac{1}{\Gamma(\alpha) \beta^\alpha} e^{-\frac{x}{\beta}} x^{\alpha-1}$; $x > 0$; $\Theta = \{(\alpha, \beta) : \alpha > 0, \beta > 0\}$.

$$f_\theta(x) = 1 \cdot \exp\left[-\frac{1}{\beta}x + (\alpha-1)\ln x + \ln \frac{1}{\Gamma(\alpha) \beta^\alpha}\right], \text{ so that}$$

$$h(x) = 1, s=2, \eta_1(\theta) = -\frac{1}{\beta}, \eta_2(\theta) = (\alpha-1), T_1(x) = x, T_2(x) = \ln x,$$

$$B(\theta) = \cancel{B(\alpha, \beta)} = B(\alpha, \beta) = -\ln\left(\frac{1}{\Gamma(\alpha) \beta^\alpha}\right) = \ln(\Gamma(\alpha) \beta^\alpha).$$

$\Rightarrow$ Gamma (α, β) belongs to the 2-par. exponential family of distr $\alpha > 0, \beta > 0$

$$\text{Ex 2 } X \sim U(0, \theta), f_\theta(x) = \frac{1}{\theta}, 0 \leq x \leq \theta = \frac{1}{\theta} I_{(0, x)} I_{(x, \theta)}$$

cannot be expressed as (*) because of the indicator fn $I_{(x, \theta)}$
 $U(0, \theta), \theta > 0$ does not belong to exponential family of distr

Note By the NFFT, $(T_1(x), \dots, T_s(x))$ is jointly sufficient for $(\theta_1, \dots, \theta_s)$, or $(\eta_1, \dots, \eta_s)$, the natural parameters

A. The s -parameter exponential family distⁿ representation in (x) can be rewritten in a canonical & (re-parameterized) form, viz.,

$$f_{\eta}(x) = h(x) \exp \left[\sum_{i=1}^s \eta_i T_i(x) - A(\eta) \right], \quad N = \{ \eta : \eta \in \mathcal{P} \} \text{ is the natural parameter space.}$$

(η_i 's are the natural parameters)

B. If there is any linear relationship among two or more $T_i(x)$'s, we can reduce the value of s to get the minimal canonical form. The associated $T_i(\eta)$'s will jointly be minimal sufficient for the corresponding θ_i 's (or, η_i 's). The natural parameter space will also change accordingly.

Ex $X \sim \text{Bin}(n, \theta)$, $\mathcal{P} = \{ \theta : 0 < \theta < 1 \}$. (This is the usual par. space).

$$f_{\theta}(x) = \binom{n}{x} \theta^x (1-\theta)^{n-x}, \quad 0 < \theta < 1$$

$$= \binom{n}{x} \exp [x \ln \theta + (n-x) \ln(1-\theta)]$$

$$= f_{\eta}(x).$$

Then, $s=2$, $h(x) = \binom{n}{x}$, $\eta_1(\theta) = \ln \theta = \eta_1$, $T_1(x) = x$, $B(\theta) = 0$
 $\eta_2(\theta) = \ln(1-\theta)$, $T_2(x) = (n-x)$, $A(\eta) = n \ln(1-\theta)$
 (but should be expressed in terms of η , and not θ).

We must get θ in terms of η , but let us do it after attaining the minimal canonical form, as given below:

$$f_{\eta}(x) = \binom{n}{x} \exp \left[\left\{ \ln \frac{\theta}{1-\theta} \right\} x + n \ln(1-\theta) \right], \text{ so now } s=1, h(x) = \binom{n}{x},$$

$$\eta = \ln \frac{\theta}{1-\theta}, \quad T(x) = x, \quad A(\eta) = n \ln(1+e^{\eta}) \quad (\text{Why?}).$$

Here it was possible to reduce $s=2$ to $s=1$, since $T_1(x) = x$ and $T_2(x) = n-x$ in the usual ~~ex~~ and canonical form had a linear relationship.

NB: (i) natural parameter space of the minimal canonical form here is $H = \{ \eta : \eta \in \mathbb{R} \}$, which is different from that of the canonical form.

(ii) The associated statistic $T(x) = x$ is a minimal sufficient statistic for θ (or, η).

C If the natural parameter space of the minimal canonical form contains an s -dimensional open rectangle, the s -parameter exponential family distribution is a full-rank s -parameter exponential family distⁿ.

eg. The natural parameter sp. of the minimal canonical form in the above example, $H = \{ \eta : \eta \in \mathbb{R} \}$ is an open interval, and hence is $\text{Bin}(n, \theta)$ is a full rank 1-par. exponential family distⁿ (n fixed).

II. (After I. Direct Approach in Lecture 10).
Indirect approach to verify whether a sufficient or, minimal sufficient statistic is complete or not using the exponential family distribution concept.

Result (R) If an s -parameter exponential family distⁿ is of full rank, then the associated minimal sufficient statistic is complete (Without proof!).

Hence, in Bin(n, θ), with a fixed n , and $\theta \in (0, 1)$, $T(x) = X$, the minimal sufficient statistic is complete!

Ex $X \sim N(\mu, \sigma^2)$, $\theta = (\mu, \sigma)^T$, $\Theta = \{(\mu, \sigma)^T, \mu \in \mathbb{R}, \sigma > 0\}$, $x \in \mathbb{R}$

$$f_{\theta}(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\},$$

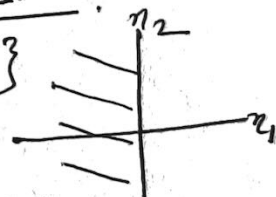
$$= \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2\sigma^2}x^2 + \frac{\mu}{\sigma^2}x - \frac{\mu^2}{2\sigma^2} - \ln \sigma\right].$$

$$\text{Finding, } h(x) = 1 \left(n, \frac{1}{\sqrt{2\pi}}\right), \quad s=2, \quad \eta_1 = \eta_1(\theta) = -\frac{1}{2\sigma^2}, \quad T_1(x) = x^2$$

$$\eta_2 = \eta_2(\theta) = \frac{\mu}{\sigma^2}, \quad T_2(x) = x \quad \text{and} \quad B(\theta) = \frac{\mu^2}{2\sigma^2} + \ln \sigma.$$

The canonical form is same except for $A(\eta) = A(\eta_1, \eta_2)$ replacing $B(\theta) = B(\mu, \sigma)$ find $A(\eta)$

The natural parameter sp. $H = \{(\eta_1, \eta_2), \eta_1 < 0, \eta_2 \in \mathbb{R}\}$ contains an open rectangle of dimension 2



No further reduction of dimension is possible to a different minimal canonical form.

Hence the associated MSS (x^2, x) is complete.

Note In most situations, we shall be given a random sample $x_1, \dots, x_n$ from f_{θ} , so that we can express $f_{\theta}(x) = \prod_{i=1}^n h(x_i) \exp\left[\sum_{i=1}^n \eta_1 T_1(x_i) + \sum_{i=1}^n \eta_2 T_2(x_i) - nA(\eta)\right]$, as the canonical/minimal canonical form.

~~But~~ In the above example - from $N(\mu, \sigma^2)$
 Then $(T_1(x), \dots, T_2(x))$ is jointly sufficient or minimal sufficient and is complete if it satisfies the condition of full-rank.
 In the above example $N(\mu, \sigma^2)$, we then say
 the MSS $T(x) = \left(\sum_{i=1}^n x_i^2, \sum_{i=1}^n x_i\right)$ is complete (and not (x^2, x) is complete!)
 $= (\sum x_i - \bar{x})^2, \sum x_i)$
 $= (\bar{x}, s^2)$

Next, Example of a non-full rank exponential family